

with transformative designs, 360-degree gimbals, 4k video, higher-value instrumentation, improved safety, autopilot and intelligent piloting mode. Commercially-suitable drones must have fully-compliant safety and regulatory standards-based designs, automated safety mode, platform and payload adaptability, full airspace awareness with auto-action like take-off, landing, mission execution and intelligent piloting, and full autonomy.

Robots in services, security and surveillance

Robotic surgery is one of the most important innovations of the medical field. A da Vinci single-port surgical system can help treat deep-seated damaged or diseased tissues or organs.

Robots like Saul are designed to fight diseases or eliminate viruses. It has been used to eradicate Ebola virus by using powerful pulses of highly energetic ultraviolet (UV) rays.

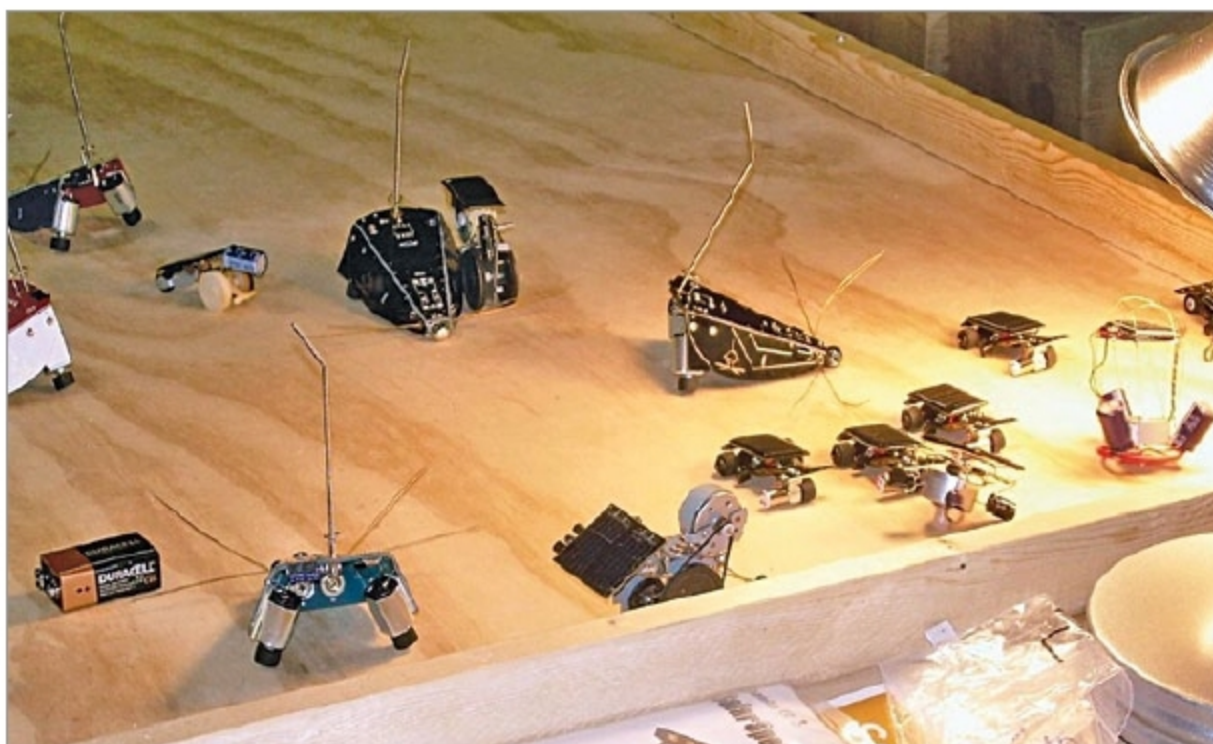
Wearable robots like Exosuit increase the wearer's strength, balance and endurance with personalised control. Such robots help enhance the muscles of the physically-disabled to help them walk and climb. These also find applications in military.

Robots like Asus Zenbo are used for home services that understand voice commands. These send reminders for daily tasks, medication schedules, doctor's appointments, detect emergencies and connect to other smart home components.

Robotics in dairy farms have become a major industry, with thousands of robotic milking parlours already been installed worldwide.

Industrial robots have gained popularity for their productivity, efficiency and quality. Collaborative robots—also called cobots—work safely alongside humans. Smart sensors are connected to internal systems to collect production data, previously inaccessible to manufacturers. This also increases cybersecurity risks for manufacturers in their processes. So, these are required to ensure safe and reliable production and to improve product compatibility.

Robots are used for assembling products, space exploration, handling dangerous materials, spray-painting, cutting and polishing, inspection of



Microbots (Credit: Flickr)

products and other processes. The number of robots used in tasks as diverse as cleaning sewers, detecting bombs, robotic lawn mowers, retail and security, agriculture and logistics will continue to grow in the coming years. Robots would eventually be used in fast food restaurants.

Microbots help emergency responders explore environments that are too small or too dangerous for humans or larger robots. Robots powered with alternate sources like solar, wind and wave energy are used in areas that are off the grid. Robots inspired by living creatures, such as DelFly Nimble, serve to develop advanced technologies that have practical applications in surveillance.

In security environments, autonomous robots are being developed to collect and react to security-related information. These carry multiple visual and thermal cameras, gas sensors, two-way communication systems and so on. The robots are also empowering law and order agencies by giving an edge to smart workplaces.

Drones to analyse surroundings and avoid obstacles

Rather than using onboard systems that increase weight and demand more battery power, software allows streaming the videos back to an object-recognition server. This enables drones to analyse the terrain and act based on certain sets of predefined instructions using a GPS-guided mode. The biggest challenge in gain-

ing commercial use of drones is the development of systems to prevent them from crashing into people or property, each other and, even more important, into manned aircraft.

Collision avoidance technology and algorithms for unmanned aerial systems (UAVs) will enable drone operators and businesses to rely on drone services. These systems fuse one or more of the following sensors to sense and avoid obstacles: vision, ultrasonic, infrared (IR), lidar, time-of-flight or monocular vision.

To overcome safety and privacy issues, there is a need to regulate the maximum height for flight, weather and time-of-day conditions for flight, and areas where flights may be prohibited such as near airports. Over the next few years, drone technology will become better in providing safer, more automated and quieter drones. Technologies that could transform future drone operations are IR cameras, GPS and lasers, hydrogen power, improved computer vision and motion planning.

Radar positioning and failsafe function

Many of the latest drones have dual Global Navigation Satellite Systems (GNSS) such as GPS and Global Navigation Satellite System (GLONASS). Drones can fly in both GNSS and non-satellite modes. Highly accurate drone navigation is important for creating 3D maps, surveying landscapes and search-and-rescue (SAR) missions. These are used to display the current